Computing the core entropy of real quadratic polynomials

notes accompanying the code in `src/CoreEntropy.jl`

1 Introduction

For $\lambda \in [2,4]$ let

$$g_\lambda(x) = \lambda x(1-x), \qquad x \in [0,1]. \tag{1}$$

The map g_λ is unimodal: it increases from 0 to the critical value $\lambda/4$ on $[0, \frac{1}{2}]$ and decreases back to 0 on $[\frac{1}{2}, 1]$. Its *core entropy* $h(\lambda)$ is the topological entropy of g_λ restricted to its Hubbard tree, the smallest invariant interval carrying the critical point and its orbit. Equivalently — and this is how it will be computed — it is the exponential growth rate of the number of laps of the iterates g_λ^n.

As λ runs from 2 to 4 the parameter crosses the real slice of the quadratic family and h rises from 0 to $\log 2$, but not smoothly. It is constant on every hyperbolic component, that is, on every interval of parameters for which an attracting cycle exists, and it changes only on the complement of their union. The graph is therefore a devil's staircase, with a plateau for each attracting cycle and a Cantor-like set of parameters on which the increase takes place. Tiozzo [4] proved that h is nevertheless continuous in the parameter, and related its values to the dimensions of sections of the Mandelbrot set.

The purpose of this note is to record the algorithm by which the staircase is computed, the conventions it uses and the control it keeps over the truncation error. Section 2 fixes the object, §3 recalls the kneading theory that turns the entropy into a root of a power series together with the piecewise-linear model that reproduces the lap counts, §4 and 5 describe and analyse the numerical scheme, and §6 collects the evidence that the implementation is correct. The code, `src/CoreEntropy.jl`, follows the same order; the two front ends contain no mathematics of their own.

2 The core entropy of a real quadratic polynomial

Everything below is stated for the logistic family, but the object it computes is usually defined for the quadratic family $f_c(z) = z^2 + c$, and the two are the same thing:

Lemma 2.1. *Let $\psi_\lambda(x) = \frac{\lambda}{2} - \lambda x$. Then $\psi_\lambda \circ g_\lambda = f_c \circ \psi_\lambda$, where*

$$c = \frac{\lambda}{2} - \frac{\lambda^2}{4} = \frac{\lambda(2-\lambda)}{4}. \tag{2}$$

The map $\lambda \mapsto c$ decreases from $c = 0$ at $\lambda = 2$ to $c = -2$ at $\lambda = 4$, with inverse $\lambda = 1 + \sqrt{1-4c}$ on this range.

Proof. Substituting $x = \frac{1}{2} - z/\lambda$ gives $x(1-x) = \frac{1}{4} - z^2/\lambda^2$, hence $\psi_\lambda(g_\lambda(x)) = \frac{\lambda}{2} - \lambda^2(\frac{1}{4} - z^2/\lambda^2) = \frac{\lambda}{2} - \frac{\lambda^2}{4} + z^2$. The derivative of (2) is $\frac{1}{2} - \frac{\lambda}{2}$, which is negative for $\lambda > 1$, so c decreases; the inverse is the positive root of $\lambda^2 - 2\lambda + 4c = 0$. □

Thus g_λ on $[0,1]$ is the quadratic family on a real interval, conjugated by an orientation-reversing affine map, and the parameter range $\lambda \in [2,4]$ corresponds to $c \in [-2, 0]$.

Definition 2.2. Let c lie in the Mandelbrot set, K_c the filled Julia set of f_c and $P_c = \overline{\{f_c^n(0) : n \geq 0\}}$ the postcritical set. The *Hubbard tree* $T_c \subset K_c$ is the smallest connected closed set containing P_c, and the *core entropy* is $h(c) = h_{\mathrm{top}}(f_c|_{T_c})$.

The Hubbard tree is the part of the Julia set that carries the postcritical dynamics: the Julia set itself may be a dendrite or worse, while T_c is a finite tree on which f_c acts as a Markov map. For real parameters the tree degenerates to an interval, and the whole theory becomes one-dimensional. Indeed, for $c \in [-2, \frac{1}{4}]$ the set $K_c \cap \mathbb{R}$ is an interval, so $T_c = [\min P_c, \max P_c]$ is a real interval; since f_c maps P_c to itself and is continuous, $f_c(T_c) \subset T_c$. For $c \in [-2, 0]$ the critical point 0 belongs to T_c and is its only turning point — a minimum, because $f_c' = 2z$ is negative to the left of 0 and

positive to the right. Passing back through the orientation reversing conjugacy of Lemma 2.1, the core becomes the convex hull of $\{\frac{1}{2}\} \cup \{g_\lambda^n(\frac{1}{2})\}$, on which g_λ is unimodal with a *maximum* at $\frac{1}{2}$. The theory below is formulated for this map; we write $h(\lambda) = h(c(\lambda))$.

Since the core is the smallest invariant interval, every orbit outside it eventually falls into it, so the entropy of g_λ on all of $[0,1]$ agrees with $h(\lambda)$; for a unimodal map the entropy is in any case determined by the itinerary of the critical point, which the core contains. Four values anchor the computation and serve as tests: $h(2) = 0$, because the critical point is fixed ($c = 0$) and the map is conjugate to $z \mapsto z^2$ on an interval; $h(1+\sqrt{5}) = 0$, the centre of the basilica component ($c = -1$), where the critical point has period 2; $h = 0$ on the whole period-doubling cascade up to the Feigenbaum parameter $\lambda_\infty \approx 3.5699457$, where an attracting cycle always captures the critical orbit; and $h(4) = \log 2$, the Chebyshev case $c = -2$, where g_4 is conjugate to the full tent map.

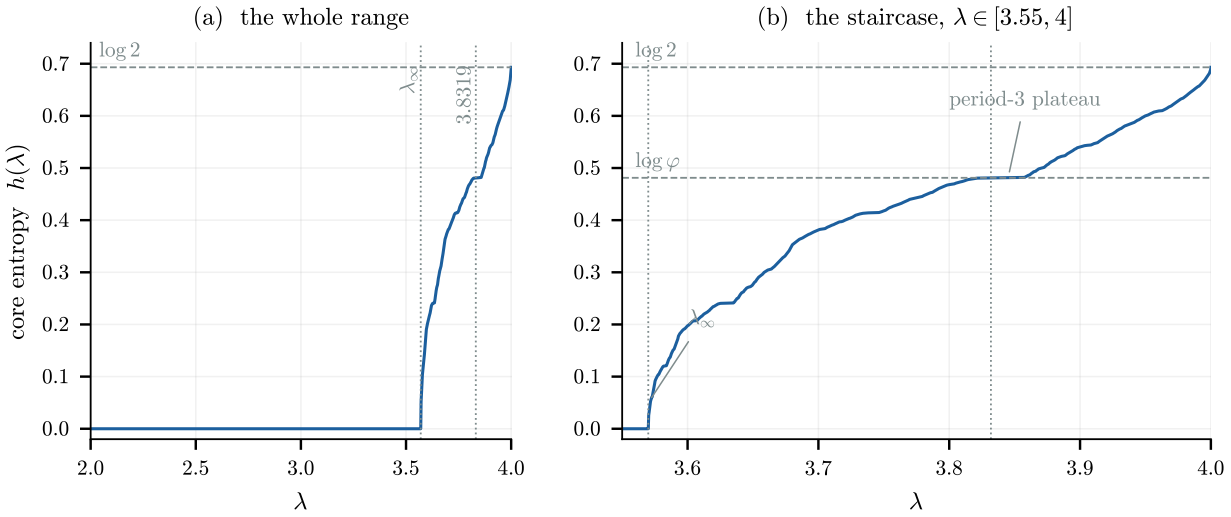


Figure 1: The core entropy of g_λ on $[2,4]$, as computed by the code described here on 28 696 parameters. (a) The whole range: h vanishes identically up to the Feigenbaum parameter λ_∞ and reaches $\log 2$ at $\lambda = 4$. (b) The staircase on $[3.55, 4]$; the dashed lines are $\log \varphi$ and $\log 2$.

Figure 1 is the output of the code on 28 696 parameters of $[2,4]$. Panel (a) is the whole range: h vanishes identically up to the Feigenbaum parameter λ_∞, where the period-doubling cascade accumulates, and then climbs to $\log 2$. Panel (b) resolves the climb. It is not a curve but a staircase — a flat step for every hyperbolic component, joined by steep pieces on which the parameter passes through the queer set — and the widest of the steps is the one at $\log \varphi$, whose length is explained in §6.

3 Kneading theory

3.1 Itineraries, laps and entropy

The combinatorial type of an orbit is recorded by the sequence of sides of the critical point it visits. Since g_λ is monotone on each side, this sequence determines everything that monotonicity can see, and in particular it determines the entropy.

Definition 3.1. For $x \in [0,1]$ the *itinerary* $\iota(x) = (\iota_0(x), \iota_1(x), \dots)$ is given by $\iota_k(x) = 0$ if $g_\lambda^k(x) < \frac{1}{2}$ (left of the critical point), $\iota_k(x) = 1$ if $g_\lambda^k(x) > \frac{1}{2}$ (right), and $\iota_k(x) = *$ if $g_\lambda^k(x) = \frac{1}{2}$. A *lap* of a continuous interval map F is a maximal interval on which F is monotone, and ℓ_n denotes the number of laps of g_λ^n.

Lemma 3.2. *For every $n \geq 1$ the laps of g_λ^n are exactly the nonempty sets*

$$I_w = \{x \in [0,1] : \iota_0(x) \cdots \iota_{n-1}(x) = w\}, \qquad w \in \{0,1\}^n,$$

and g_λ^n is monotone on each of them. Hence ℓ_n equals the number of admissible *words of length n, the words that occur as itineraries.*

Proof. For $n = 1$ this is the definition of the two laps $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$. Suppose the assertion holds for n and let I_w be a lap of g^n; on it g^n is monotone, so g^{n+1} is monotone exactly on the two pieces where $g^n(x) < \frac{1}{2}$ and $g^n(x) > \frac{1}{2}$, namely I_{w0} and I_{w1}. These are intervals because g^n is monotone on I_w, and every lap of g^{n+1} arises from exactly one lap of g^n in this way. A nonempty I_w is maximal, since enlarging it would add points whose itinerary differs from w in the first n symbols. □

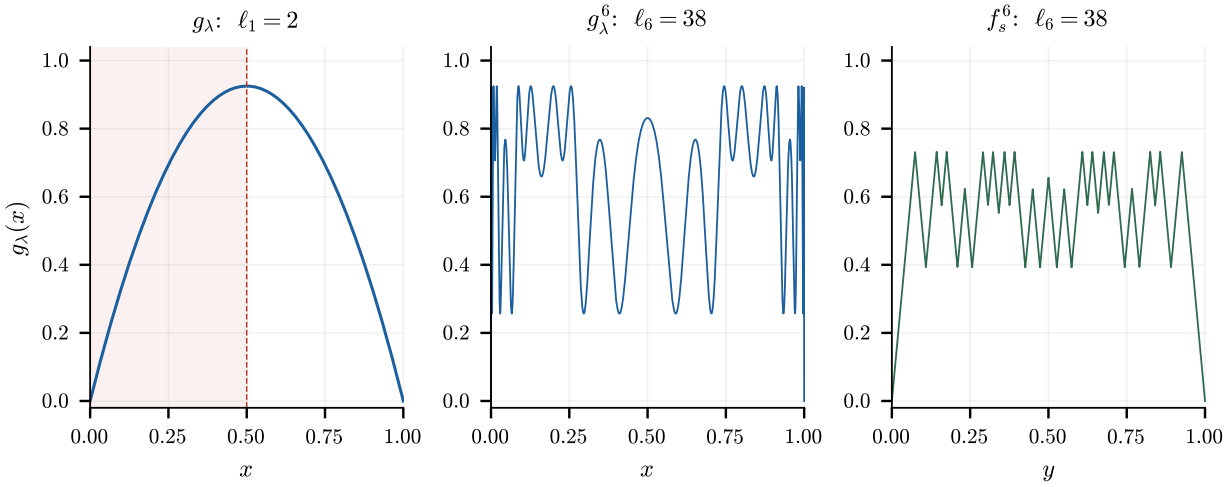


Figure 2: Laps at $\lambda = 3.7$. (a) g_λ, with its two laps shaded alternately. (b) the sixth iterate g_λ^6, drawn on a grid of 3000 points: its 38 laps are the oscillations of the graph. (c) the sixth iterate of the tent map of §3.5, of slope $e^{h(\lambda)} = 1.4648$, which has the same 38.

Figure 2 illustrates the lemma at $\lambda = 3.7$. Panel (a) shows g_λ with its two laps; panel (b) the sixth iterate on a fine grid, where the laps are the oscillations and there are 38 of them; panel (c) the sixth iterate of the tent map that §3.5 associates with the same parameter, which has the same 38. The counts for $n = 1, 2, 4, 6, 8$ are 2, 4, 14, 38, 90. The first values look like 2^n, and to begin with they are, since a lap of g^n has at most two preimage-laps under g; the count falls away from 2^n only once the preimage tree is pruned, which happens because a point $y > \lambda/4$, the maximum of g_λ, has no real preimage at all. The growth rate at $\lambda = 3.7$ is $e^h = e^{0.3817} \approx 1.46$, far below 2: the interval is not folded everywhere.

The lemma turns a dynamical quantity into a combinatorial one, and the entropy is the exponential growth rate of the count:

Theorem 3.3 (Misiurewicz–Szlenk [3])**.** *For a piecewise monotone interval map,* $h = \lim_{n\to\infty} \frac{1}{n} \log \ell_n$.

The limit exists because $\ell_{m+n} \le \ell_m \ell_n$ — cutting a lap of g^{m+n} by g^m gives at most ℓ_m pieces, each monotone for g^n — so $\log \ell_n$ is subadditive and the limit equals the infimum. Since g_λ has two laps, $\ell_n \le 2^n$ and therefore

$$0 \le h(\lambda) \le \log 2. \tag{3}$$

For the entropy one only needs the growth rate, but ℓ_n itself grows like e^{hn}, so enumerating laps is exponentially expensive and useless beyond small n. The kneading determinant computes the growth rate from a single sequence of signs.

3.2 The kneading sequence and the determinant

Definition 3.4. Let $x_k = g_\lambda^k(\frac{1}{2})$. The *kneading sequence* $\nu = \nu_1 \nu_2 \cdots$ is the itinerary of the critical value: ν_k is the symbol of x_k. If $x_k = \frac{1}{2}$ for some $k \ge 1$, that is, if the critical point is periodic of period k, then $\nu_k = *$ and the sequence is declared periodic of period k, with $*$ read as 1 in all later positions.

The sequence ν is the complete combinatorial invariant of the critical orbit, and by the results of [2] it determines the entropy. When the critical point is periodic the itinerary is undefined at the return, and the symbol has to be given a value. We read $*$ as 1, so that $s_k = -1$ there. This is one of the two one-sided limits of the kneading sequence as the parameter approaches such a value, and since h is continuous in the parameter [4], both limits give the same entropy; the choice only fixes which of the two descriptions is written down.

Definition 3.5. The *sign sequence* is $s_k = \operatorname{sgn} g'_\lambda(x_k)$. Since $g'_\lambda = \lambda(1-2x)$, one has $s_k = +1$ when $x_k < \frac{1}{2}$ and $s_k = -1$ when $x_k > \frac{1}{2}$; at a critical hit $x_k = \frac{1}{2}$ we set $s_k = -1$, which is exactly the rule "read $*$ as 1", the symbol of the decreasing branch. With $\varepsilon_0 = 1$ put

$$\varepsilon_k = s_1 s_2 \cdots s_k \qquad (k \geq 1). \tag{4}$$

The sign s_k records which branch of g_λ the orbit is on at time k, and the product ε_k records the orientation of g_λ^k on the lap whose itinerary is the first k symbols of ν: the map is increasing there when $\varepsilon_k = +1$ and decreasing when $\varepsilon_k = -1$.

Definition 3.6. The *kneading determinant* of ν is the power series

$$D(t) = \sum_{k \geq 0} \varepsilon_k t^k. \tag{5}$$

Since $|\varepsilon_k| = 1$ for every k, D converges absolutely on the disc $|t| < 1$, is continuous there, and $D(0) = 1$.

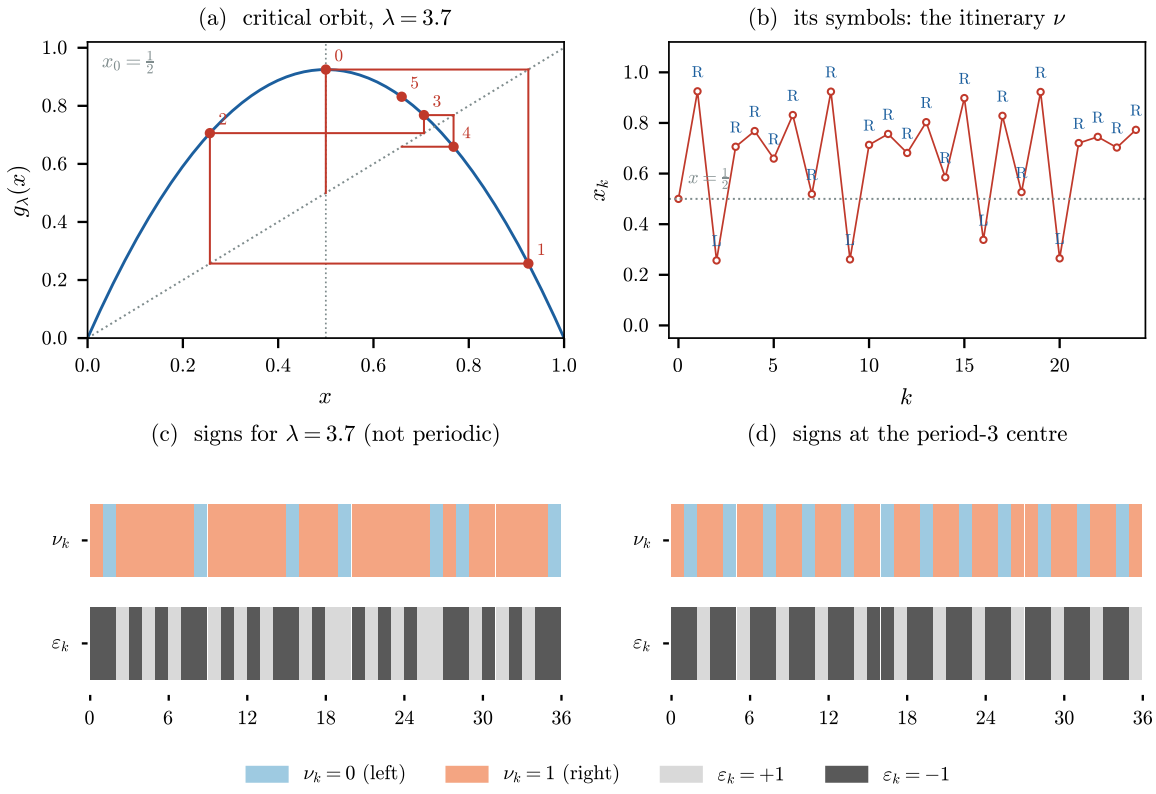


Figure 3: The kneading data. (a)–(c) at $\lambda = 3.7$: the cobweb of the critical orbit; the same orbit against time with the symbol of each point written above it; and the itinerary ν together with the partial products ε_k. (d) at the period-3 centre $\lambda = 3.8319$, where the critical point returns to $\frac{1}{2}$ and both sequences are periodic: $(1, 0, *)$ repeated, with $*$ read as 1. In panel (b) the letters are the symbols: L is 0, R is 1.

Figure 3 shows what the data look like for one parameter. Panel (a) is the cobweb of the critical orbit on the graph of g_λ: from $x_0 = \frac{1}{2}$ the orbit jumps to the critical value and then wanders. Panel (b) plots the same numbers against time and records above each point the symbol of the side it lies on; that string is the itinerary ν. Panel (c) displays it as a strip of cells together with the partial products ε_k. Each is a sequence of signs and nothing more, and the content of §3 is that these signs determine the entropy. Panel (d) repeats the exercise at the period-3 centre, where the critical point returns to $\frac{1}{2}$ after three steps: the string is then periodic, $(1, 0, *)$ repeated with $*$ read as 1, and this periodicity of the data is exactly what makes the determinant rational in §3.4.

3.3 The zero of the determinant is the growth rate

Theorem 3.7 (Milnor–Thurston [2]; see also [1], Ch. II)**.** *Let h be the topological entropy of g_λ. If $h > 0$ then $D(e^{-h}) = 0$ and $D(t) > 0$ for $0 < t < e^{-h}$. If $h = 0$ then $D(t) > 0$ for all $t \in (0, 1)$.*

Consequently e^{-h} is the smallest zero of D in $(0, 1)$, and

$$h(\lambda) = \log \frac{1}{t_0}, \qquad t_0 = \min\{t \in (0, 1) : D(t) = 0\}, \tag{6}$$

with the convention that no such t_0 exists exactly when $h = 0$. By (3), $t_0 \geq \frac{1}{2}$ whenever $h > 0$. This is the identity on which the whole computation rests: it replaces a limit of lap counts by the root of a series with coefficients ± 1. Figure 5(a) shows such a zero.

Remark 3.8 (why a zero of D is a growth rate)**.** The reciprocal $1/D$ is the kneading zeta function of the map: its coefficients count periodic orbits with signs determined by their itineraries [2]. For a piecewise monotone map the number of periodic orbits grows like e^{hn}, so $1/D$ has radius of convergence e^{-h}. By Pringsheim's theorem a series with nonnegative coefficients is singular on the positive axis; here $1/D$ is singular at $t = e^{-h}$, so D vanishes there. The second half of the theorem, the absence of zeros below e^{-h}, says that this is the *first* singularity and is what makes the smallest zero the correct one to take. We use the theorem as a black box and test it in §6.

3.4 The postcritically finite case, exactly

When the critical orbit is periodic of period k the sign sequence is k-periodic, so ε is periodic with period k or $2k$ and D is a rational function $D = P(t)/(1 - t^p)$ whose numerator P has degree $p - 1$ and integer coefficients. The zeros of D are then the zeros of P, and t_0 is the smallest of them in $(0, \frac{1}{2})$: an algebraic number that can be computed exactly, with no truncation at all. Three such parameters serve as tests.

Example 3.9 ($\lambda = 4$, the Chebyshev case)**.** Here $x_1 = 1$ and $x_k = 0$ for $k \geq 2$, so $s = (-1, +1, +1, \dots)$ and $\varepsilon_k = -1$ for every $k \geq 1$. Summing the geometric tail,

$$D(t) = 1 - \frac{t}{1 - t} = \frac{1 - 2t}{1 - t},$$

whose smallest positive zero is $t_0 = \frac{1}{2}$; hence $h = \log 2$, as it must be for a map conjugate to the tent map.

Example 3.10 ($\lambda = 1 + \sqrt{5}$, the basilica)**.** The critical orbit is $\frac{1}{2} \mapsto \frac{\lambda}{4} \mapsto \frac{1}{2}$, so $\nu = 1*$ and every $s_k = -1$. Then $D(t) = \frac{1}{1+t}$, which has no zero in $(0, 1)$, and the theorem correctly reports $h = 0$: the critical orbit is attracted to a 2-cycle and the core supports no exponential complexity.

Example 3.11 (the period-3 centre)**.** Let $c_3 = -1.7548776662\dots$ be the centre of the period-3 component and $\lambda_3 = 1 + \sqrt{1 - 4c_3}$. The critical orbit has period 3 with itinerary $(1, 0, *)$, so $s = (-1, +1, -1)$ repeated and ε has period 3 with $\varepsilon = (1, -1, -1)$. Hence

$$D(t) = \frac{1 - t - t^2}{1 - t^3},$$

whose smallest zero in $(0, 1)$ is $t_0 = 1/\varphi$ with $\varphi = \frac{1+\sqrt{5}}{2}$, so $h(\lambda_3) = \log \varphi = 0.481211825\dots$.

In both cases the entropy is settled by a polynomial rather than a series: the determinant is rational and its smallest zero can be read off exactly, with no truncation at all.

3.5 The piecewise-linear model

The lap counts are not special to the quadratic family. Every piecewise monotone map with growth number $s > 1$ is topologically semi-conjugate to one whose slope is $\pm s$ everywhere, and for the logistic family that model is the tent map of slope $s = e^{h(\lambda)} = 1/t_0$:

Theorem 3.12 (Milnor–Thurston [2], Theorem 7.4)**.** *Let f be piecewise monotone with growth number $s > 1$. Then there is a continuous, nondecreasing, surjective A from the interval of f onto $[0,1]$, and a piecewise linear $F : [0,1] \to [0,1]$ with $|F'| = s$ everywhere, such that*

$$F(A(x)) = A(f(x)) \qquad \text{for every } x.$$

The growth number of F is s.

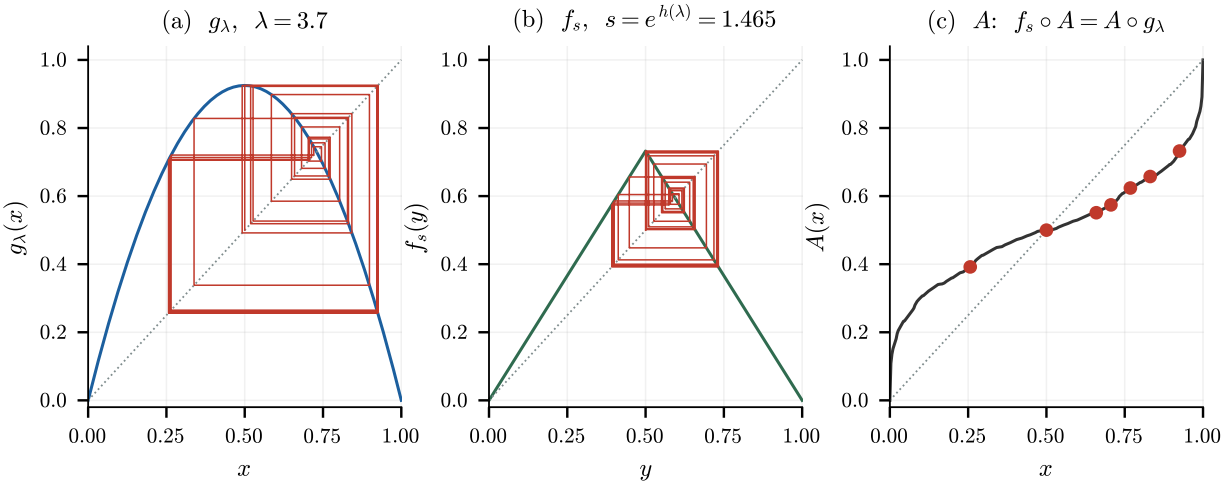


Figure 4: The three objects at $\lambda = 3.7$. (a) g_λ, with the cobweb of the critical orbit. (b) the tent map f_s of slope $s = e^{h(\lambda)} = 1.4648$, with its own cobweb. (c) the map A of Theorem 3.12, on which the iterates of the critical orbit of f_s (abscissae) sit at the corresponding iterates of g_λ (ordinates).

The map A is not a conjugacy. It may collapse intervals of the interval of g_λ to points — in particular the basin of an attracting cycle goes to a cycle of f_s — and the arrow does not reverse: a tent map has no attracting cycle, while g_λ may have one. Figure 4 shows the three objects; panel (c) is computed by following the itinerary of x and pulling a seed back through the branches of f_s, which contracts at rate s^{-n}.

What this gives the computation is a second way to see the lap numbers. Since $F(A(x)) = A(f(x))$, the map A carries each lap of g_λ^n into a lap of f_s^n, so $\ell(f_s^n) \le \ell(g_\lambda^n)$; both grow like s^n, and at $\lambda = 3.7$ they agree at every n we computed, as panels (b) and (c) of Figure 2 show for $n = 6$. The growth number s itself is what §4 computes, as $1/t_0$.

4 The algorithm

The program takes a parameter $\lambda \in [2,4]$ and returns $h(\lambda)$ by locating the smallest zero of the truncated determinant. Three constants govern the truncation: the initial order $N_0 = 800$, the maximal order $N_{\max} = 51200$, and the tolerance $\rho = 25$ whose role is explained in §5. For an order N the series (5) is truncated to $D_N(t) = \sum_{k \le N} \varepsilon_k t^k$.

Step 1: the orbit and its signs. Starting from $x = \frac{1}{2}$, for $k = 1, \dots, N$ one computes

$$x \longleftarrow \lambda x(1-x), \qquad s_k \longleftarrow \begin{cases} +1, & x < \frac{1}{2}, \\ -1, & x \ge \frac{1}{2}, \end{cases} \qquad \varepsilon_k \longleftarrow \varepsilon_{k-1} s_k.$$

The comparison is the sign of $g'_\lambda(x_k)$, and the rule $s_k = -1$ at $x = \frac{1}{2}$ implements Definition 3.4. The rule is self-consistent: if $x_k = \frac{1}{2}$ then $x_{k+1} = \lambda/4 = x_1$, so the orbit repeats from the beginning and the computed sign sequence is periodic from that point on, exactly as the convention requires.

Step 2: evaluation. The truncated determinant is evaluated by Horner's rule,

$$D_N(t) = \varepsilon_0 + t\bigl(\varepsilon_1 + t(\varepsilon_2 + \cdots + t\varepsilon_N)\cdots\bigr),$$

which costs N multiplications and additions, in one pass over the ε_k and without forming the powers t^k separately.

Step 3: the smallest zero. D_N is continuous on $[0,1]$ with $D_N(0) = 1$, and by (3) the zero of interest lies in $[\frac{1}{2}, 1]$; the search is confined to $[0.45,\, t_{\max}]$ with $t_{\max} = 1 - \rho/N$, the lower end for safety and the upper end because the truncation is unreliable as $t \to 1$ (§5). A bracket is found in one of two ways. If the previous parameter had a zero t_{prev}, one tries $[t_{\mathrm{prev}} - \delta, t_{\mathrm{prev}} + \delta]$ with $\delta = 10^{-4}$, multiplying δ by 4 (up to fourteen times) until $D_N > 0$ at the left end and $D_N < 0$ at the right end; since the zero moves continuously with the parameter, this almost always succeeds and costs a handful of evaluations. If it fails, 80 equally spaced points of $[0.45, t_{\max}]$ are evaluated and the first sign change from + to − is taken. The bracket is then refined by bisection until the interval stops shrinking at machine precision.

Step 4: acceptance, doubling, and the case of zero entropy. With t_0 in hand, $h = -\log t_0$. The zero is accepted if $hN \geq \rho = 25$ and N is doubled otherwise, up to $N_{\max}$; if no sign change exists at all, h is reported as 0 after one precautionary increase of N, from 800 to 3200. Both rules exist to keep the truncation error under control, and both are justified in §5.

It may help to follow one parameter through the four steps. For $\lambda = 3.7$ the critical orbit begins 0.5, 0.925, 0.2567, 0.7061, …; its signs are those of Figure 3(b) and its partial products those of panel (c). With $N = 800$ the search runs over $[0.45, 0.9688]$, the first sign change is at $t_0 = 0.6827$, and $h = -\log t_0 = 0.38172$ — the value listed for this parameter in the data set.

A Horner evaluation costs $O(N)$ and a parameter needs of order fifteen of them once the bracket is inherited, so the sweep of 28 696 parameters used for the published data takes about 1.5 s on a single core, the largest order reached being $N = 3200$.

5 Error control

5.1 Truncation

Because $|\varepsilon_k| = 1$ for every k, the tail of (5) is dominated by a geometric series, with no information about the orbit beyond the coefficients:

$$|D(t) - D_N(t)| \leq \sum_{k>N} t^k = \frac{t^{N+1}}{1-t}, \qquad 0 < t < 1. \tag{7}$$

This is the only a priori error in the method, and the thresholds of §4 are chosen to control it.

Figure 5 tests both halves of the analysis. Panel (a) shows D_N for $\lambda = 3.7$ with its zero marked, and shades the region $t > 1 - \rho/N$ in which nothing is trusted. Panel (b) plots the distance between h_N and its limit against hN for four parameters. The four curves collapse onto one: the error is governed by $t_0^N = e^{-hN}$ and by nothing else. The dashed line at $hN = 25$ is the acceptance threshold: by then the error has fallen to about 10^{-10}, and a few more units of hN would only push it into the roundoff floor. That is why the value 25 was chosen rather than, say, 10 or 100. It is worth stressing that (7) holds whatever the computed signs are, so it also covers the case in which the orbit itself is inaccurate (see §5.4).

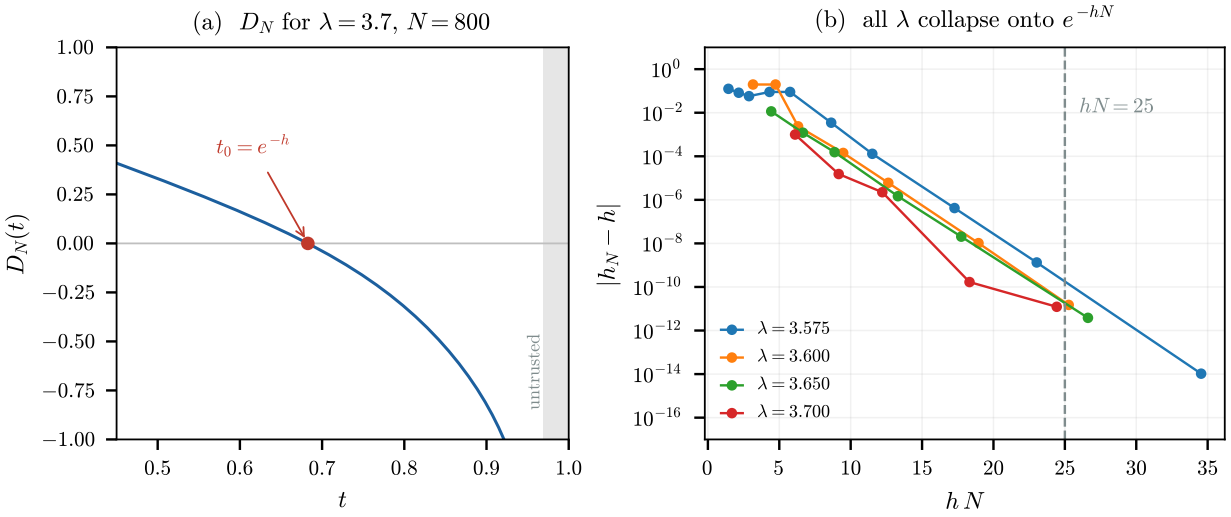


Figure 5: The zero and the truncation error. (a) $D_N(t)$ for $\lambda = 3.7$ and $N = 800$; its smallest zero is $t_0 = e^{-h}$, and beyond $t = 1 - \rho/N$ (shaded) the truncation is no longer controlled. (b) $|h_N - h|$ against hN for four parameters: the curves collapse, which is the empirical content of (7); the dashed line is the acceptance threshold $hN = 25$.

5.2 The acceptance criterion

Let $t_0 = e^{-h}$ be the zero of D and suppose $hN \geq \rho$. Since $1 - e^{-h} \geq h/2$ for $0 < h \leq 1$, the error of the truncated determinant at the true zero is at most

$$|D_N(t_0)| = |D(t_0) - D_N(t_0)| \leq \frac{e^{-h(N+1)}}{1 - e^{-h}} \leq \frac{2e^{-25}}{h},$$

which is at most $3 \cdot 10^{-9}$ when $h \geq 10^{-2}$, and far smaller for larger h. If $D'(t_0) \neq 0$, the zero of D_N differs from t_0 by at most this quantity divided by $|D'(t_0)|$. The code does not estimate D', so what is rigorously controlled is D rather than h itself; in practice the zeros are simple with $|D'|$ of order one, and the exact tests of §6 show errors at the level of roundoff.

5.3 The resolution floor

If there is no sign change in $[0.45, 1 - \rho/N]$, then either $h = 0$ or $0 < h < \rho/N$, and the two alternatives cannot be separated at order N. After raising N to 3200 the code reports 0, so a zero entropy is certified only down to $25/3200 \approx 0.008$. The effect on the data is confined to the foot of the staircase: of the 28 696 parameters, exactly one ($\lambda = 3.5699733$, a few 10^{-5} above λ_∞) is reported as 0 while its entropy is 0.0075. Recomputing the whole sweep with $N_0 = 6400$ changes no other value by more than 10^{-8}, which is the sense in which the floor is the only limitation of the default settings.

5.4 The floating-point orbit

The signs are read off a double-precision orbit, whose error grows like $e^{\Lambda k}$ with Λ the Lyapunov exponent, so the itinerary is reliable only up to $k \approx 16 \log 10/\Lambda$. Two mechanisms keep this harmless. On a hyperbolic component the critical orbit converges to the attracting cycle and, in floating point, reaches it exactly after finitely many steps; the computed sign sequence is then the true itinerary forever, and this covers every plateau of h. And when h is small the parameter is close to a parabolic or Feigenbaum value, Λ is small, the horizon is far, and the criterion $hN \geq \rho$ asks for precisely the large N that this accuracy permits. Beyond the horizon the tail bound (7) makes the unknown terms harmless, since their total weight is at most $t^N/(1 - t)$.

5.5 Degenerate zeros

The scan looks for a change of sign, so a smallest zero of even order would be passed over and a larger value of h reported. No such case occurred in any test, and in the postcritically finite case D

is a rational function whose zeros are generically simple. This is the one structural assumption of the search; it is recorded for completeness.

6 Validation

6.1 Parameters with known entropy

Table 1 compares the output with the exact values derived in §3. The first three are the cases of vanishing entropy, in which Theorem 3.7(ii) applies; the last two are Examples 3.9 and 3.11. The agreement is at the level of roundoff.

λ	c	$h(\lambda)$ expected	computed	|error|
2	0	0 (critical point fixed)	0.000000000000	0
3	$-3/4$	0 (parabolic fixed point)	0.000000000000	0
$1+\sqrt{5}$	-1	0 (basilica centre)	0.000000000000	0
3.8318740553	-1.7548777	$\log\varphi$	0.481211825060	$5.6\cdot10^{-17}$
4	-2	$\log 2$	0.693147180560	$2.2\cdot10^{-16}$

Table 1: Exact tests.

6.2 An independent count of laps

Theorem 3.3 can be tested without any kneading theory. Iterating the inverse branches of g_λ counts the solutions of $g_\lambda^n(x)=\frac{1}{2}$, a number that grows at the same exponential rate as ℓ_n; comparing its finite-n growth rate with the value produced by the determinant checks the whole chain of §3 independently. Table 2 shows the outcome, and Figure 6 the same thing graphically. The discrepancy decreases with n and is largest where h is small, because the subexponential corrections to the growth are then relatively more important.

λ	n	number of solutions	growth rate over 4 steps	$h(\lambda)$
3.62	26	3 068	0.234676	0.234370
3.72	26	67 872	0.399328	0.399570
3.83	26	392 612	0.481104	0.481212
3.95	25	4 265 466	0.600110	0.600143
4.00	22	4 194 244	0.693144	0.693147

Table 2: Independent check by counting solutions of $g^n(x)=\frac{1}{2}$.

The slopes in the figure are read off over the range in which the counts are still far from the roundoff floor; where they agree, the two computations of h agree.

6.3 An exact plateau

The most conspicuous feature of the graph is the step at $h=\log\varphi$, and it is not a numerical accident. At $\lambda=3.845$ the computed sign sequence is exactly 12-periodic, so $D=P(t)/(1-t^{12})$ with P of degree 11 and integer coefficients; P is divisible by t^2+t-1, hence $D(1/\varphi)=0$ exactly and, since no smaller zero exists, $h=\log\varphi$. At $\lambda=3.85$, 3.855 and 3.8565 a 60-digit evaluation of the truncated series gives $|D(1/\varphi)|<10^{-60}$, that is, zero to the precision of φ itself. The step runs from the saddle-node $\lambda=1+\sqrt{8}=3.8284271$ to $\lambda\approx3.8568$, the real extent of the period-3 copy of the Mandelbrot set, before h rises again.

Figure 7 places the step against the bifurcation diagram. The plateau in h coincides with the period-3 window, and it ends only where the window's own structure ends, at $\lambda\approx3.8568$; between the two the entropy does not move at all, even though the attracting cycle doubles repeatedly along the way.

7 Concluding remarks

Two things are proved along the way and one is assumed. The reduction of the entropy to the growth rate of lap numbers is a theorem of Misiurewicz and Szlenk, and the identification of that growth

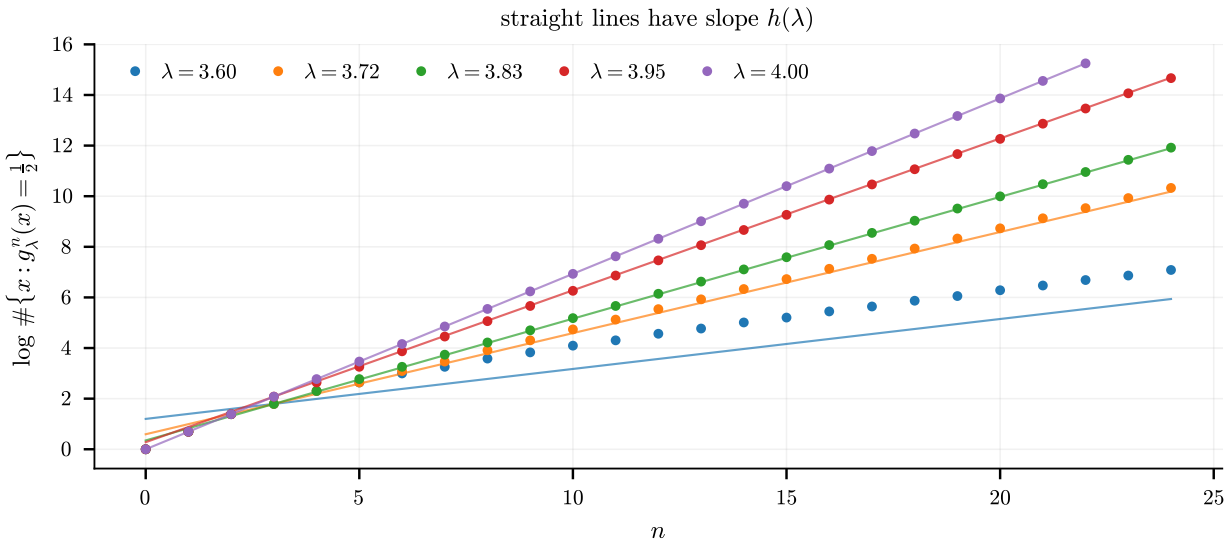


Figure 6: The independent check of Theorem 3.3: the logarithm of the number of solutions of $g_\lambda^n(x) = \frac{1}{2}$ against n, for five parameters. The superposed straight lines have slope $h(\lambda)$ as computed from the determinant.

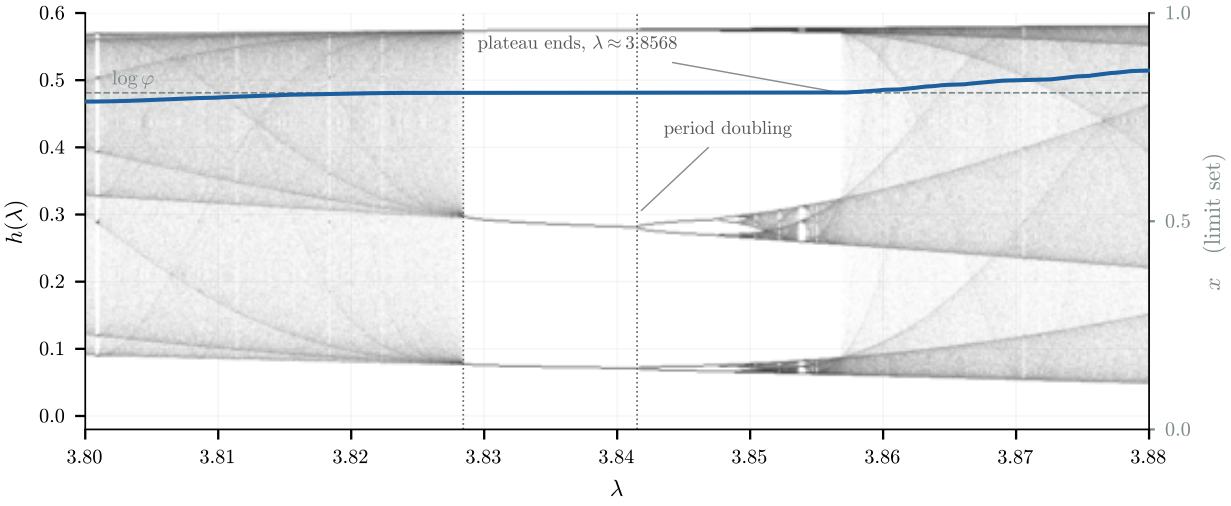


Figure 7: The plateau at $\log \varphi$ against the bifurcation diagram of the same parameters. The step in h coincides with the period-3 window and is exactly flat from the saddle-node at 3.8284 out to $\lambda \approx 3.8568$.

rate with the reciprocal of the smallest zero of the kneading determinant is a theorem of Milnor and Thurston; the truncation bound (7) is elementary and is what makes the scheme trustworthy, since it converts a series that cannot be summed into a criterion that can be checked at every parameter. What is assumed is that the smallest zero of the truncated determinant is simple, so that the search for a sign change finds it; the assumption is stated in §5 and never failed. Nothing in the method is special to the logistic family except the bound $h \le \log 2$ that places the zero above $\tfrac{1}{2}$: any unimodal family with a computable kneading sequence can be treated the same way, once the orbit and its signs are supplied.

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